Mathematics for Social Scientists

Part 2: Introduction to Analysis
Integrals

Carlos Gueiros

University of Mannheim

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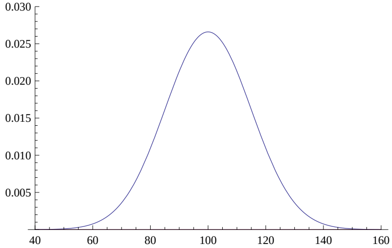
Motivation (Statistics)

- Probability density functions (p.d.f.) are fundamental in statistics.
- To get the probability of a *range* of outcomes, we need the **area under the p.d.f.**.
- Example: IQ $\sim \mathcal{N}(100, 15^2)$. What is $P(X > 120)$?

$$P(X > 120) = \int_{120}^{\infty} f_X(x)\, dx \;\approx\; 0.0912$$



Motivation: Policy & Demography

- **Political Science (mobilization impact over time).** Net effect on $[t_0, t_1]$ is the accumulated *rate* difference:

$$\text{Net effect} = \int_{t_0}^{t_1} \Big(r_{\text{policy}}(t) - r_{\text{cf}}(t) \Big)\,\mathrm{d}t = \Big[\, y_{\text{policy}}(t) - y_{\text{cf}}(t) \Big]_{t_0}^{t_1}$$

  where r is a flow/rate (e.g., *protest participants per hour*, voters contacted per day) and $y'(t) = r(t)$.

- **Sociology (demography & exposure).** Totals from time-varying rates:

$$\text{Total migrants} = \int_0^T m(t)\,\mathrm{d}t$$

  Person-time and cumulative exposure (dose):

$$\text{Person-time} = \int_0^T N(t)\,\mathrm{d}t, \qquad \text{Cumulative exposure} = \int_0^T c(t)\,\mathrm{d}t$$

  with $N(t)$ the population under observation and $c(t)$ a concentration (pollution/media).

Motivation: Psychology & Behavioral Time Courses

- **Area-under-the-curve (AUC).** Summarize response trajectories by total response:

$$\mathrm{AUC} = \int_{t_0}^{t_1} r(t)\,\mathrm{d}t \quad \text{(or above baseline: } \int_{t_0}^{t_1} [r(t) - r_0]\,\mathrm{d}t\text{)}$$

  where $r(t)$ might be cortisol/hormone level, pupil dilation, or BOLD/EEG signal.

- **Learning/engagement curves.** Total engagement over a session:

$$\text{Total engagement} = \int_0^T e(t)\,\mathrm{d}t$$

  (e.g., attention/accuracy over time in a task, adherence in interventions, app-usage intensity).

Common thread: integrate *rates/flows* to get *totals/levels*.

Areas, Sums, Integrals

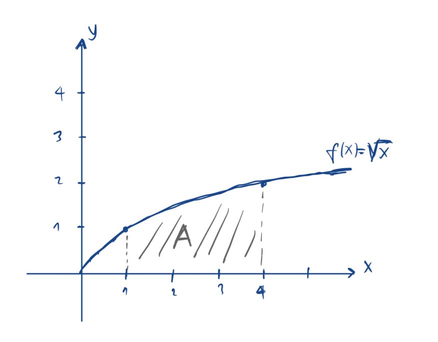
Intuition: Areas, Sums, Integrals

We want to have a neat way to calculate the area that lies ”under the function.”

Consider the following example:

Example.

Calculate the area that is under $f(x) = \sqrt{x}$ from 1 to 4:



Intuition: Integrals and Areas

We can approximate the area:

1. Divide the x-axis into equally spaced intervals,
2. construct rectangles (Rectangle areas are easy to calculate),
3. add rectangle areas.

Figures next show two different ways of doing it, with $n = 3$ disjoint intervals:

Intuition: Areas, Sums, Integrals

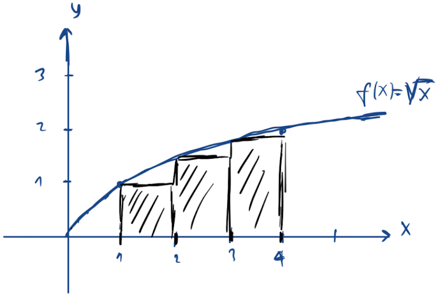


Figure: $h\sum_{i=1}^{n} f(a+ih)$
(Left Riemann Integral)

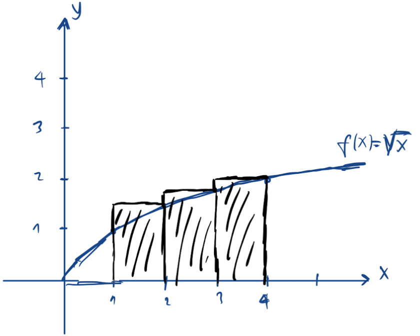


Figure: $h\sum_{i=0}^{n-1} f(a+ih)$
(Right Riemann Integral)

- Here: h is the width of each interval, that is, it is the width of the rectangle.
- The two different ways: The left (right) Riemann Integral starts from the leftest (rightest) point of the interval to measure the length of each rectangle.

Intuition: Areas, Sums, Integrals

- **Remark:** The more rectangles we have (the higher the n), the better we can approximate the area.
- Let us assume we want to calculate the area of $f(x)$ over the interval $[a, b]$. (In our example, $a = 1$ and $b = 4$)
- Then, if we construct n different intervals:

  $h = \frac{b-a}{n}$ is the width of each interval, i.e. it is the *width* of the rectangle,

  $f(a + ih)$ is the *length* of each rectangle,

  $hf(a + ih)$ is the *area* of each rectangle,

  $h\sum_{i=1}^{n} f(a + ih)$ is the *summation of all areas*, if we take select the "left" integration, and is called **Left Riemann Integral**,

  $h\sum_{i=0}^{n-1} f(a + ih)$ is the *summation of all areas*, if we take select the "right" integration, and is called **Right Riemann Integral**.

Definite Integrals,
Antiderivatives,
Fundamental Theorem of Calculus

Definite Integrals

For a continuous function $f(x)$ bounded by a and b, define the following limits for the left and right Riemann integrals:

$$S_{left} = \lim_{h \to 0} h \sum_{i=0}^{n-1} f(a + ih)$$

$$S_{right} = \lim_{h \to 0} h \sum_{i=1}^{n} f(a + ih)$$

where n is the number of bars, h is the width of the bars (and bins), and nh is required to cover the domain of the function, $b - a$.

Then, we have:

$$S_{left} = S_{right} = R$$

because of the limit.

Definite Integrals

Definition 1 (Definite Integral).

Assume $f(x)$ and the definitions are as given in the previous slide. Then the **definite integral** is given by:

$$R = \int_a^b f(x)\, dx$$

- $\int$ symbol looks similar to an "S" to remind that it is a special kind of summation,
- Placement of a and b indicates *the lower* and *upper limits* of the definite integral, i.e. boundaries for where the summation is taking place,
- $f(x)$ in the above expression is called the *"integrand"*,
- dx is a reminder that we are taking the sum over *infinitesimal* values of x,
- The integral here is called *"definite"* because limits on the integration are defined, i.e. they have specific values such as a and b here.

Definite Integrals II

An Example:
Suppose we want to evaluate the function $f(x) = x^2$ over the domain $[0, 1]$.

First divide the interval up into h slices. We have h slices of $1/h$ width, since the interval is 1 wide.

The area under $f(x) = x^2$ is given by the limit of the right Riemann integral:

$$
\begin{aligned}
R = \lim_{h\to\infty} \sum_{i=1}^{h} \frac{1}{h} f(x) &= \lim_{h\to\infty} \sum_{i=1}^{h} \frac{1}{h} (i/h)^2 \\
&= \lim_{h\to\infty} \frac{1}{h^3} \sum_{i=1}^{h} i^2 = \lim_{h\to\infty} \frac{1}{h^3} \frac{h(h+1)(2h+1)}{6}
\end{aligned}
$$

Definite Integrals III

$$= \lim_{h\to\infty} \frac{1}{6}\left(2 + \frac{3}{h} + \frac{1}{h^2}\right) = \frac{1}{3}$$

where we use the following facts:

$$\sum_{x=1}^{n} x^2 = \frac{n(n+1)(2n+1)}{6} \qquad \sum_{x=1}^{n} x = \frac{n(n+1)}{2}$$

Antiderivatives

In simple terms, an *"Antiderivative"* is the opposite of a derivative.

Definition 2 (Antiderivative).
Let $f(x)$ be a function, with the derivative $f'(x)$.
Let $f'(x) = g(x)$. Then the *antiderivative* of $g(x)$ is $G(x)$, where $G'(x) = g(x)$.

Examples:

- An antiderivative of $f(x) = x^2$ is $F(x) = \frac{1}{3}x^3 + c$
- An antiderivative of $g(x) = x^n$ is $G(x) = \frac{1}{n+1}x^{n+1} + c$
- An antiderivative of $h(x) = e^{2x}$ is $H(x) = \frac{1}{2}e^{2x} + c$
- An antiderivative of $p(x) = \frac{1}{x}$ is $P(x) = ln|x| + c$

Fundamental Theorem of Calculus

Definition 3 ((Second) Fundamental Theorem of Calculus).

Let f be a real-valued function on a closed interval $[a, b]$ and F a continuous function on $[a, b]$, which is an antiderivative of f on (a, b): $F'(x) = f(x)$.
If f is Riemann integrable on $[a, b]$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

Fundamental Theorem of Calculus: Discussion

The theorem is extremely useful (*fundamental!*).

What it says, in simple terms: differentiation and integration are opposite operations. The integral of f from a to b is the antiderivative at b minus the antiderivative at a.

Example.
Coming back to our previous example, the integral of $f(x) = x^2$ on the interval $[0, 1]$ is:

$$\int_0^1 x^2 \, dx = \left[\tfrac{1}{3}x^3\right]_0^1 = \tfrac{1}{3}(1^3) - \tfrac{1}{3}(0^3) = \tfrac{1}{3}.$$

Example.
Another example:
$\int_1^4 x \, dx = \left[\frac{1}{2}x^2\right]_1^4 = 7.5.$

Properties of Definite Integrals

- *Constants*
  $\int_a^b kf(x)dx = k\int_a^b f(x)dx$
- *Additive Property*
  $\int_a^b (f(x) + g(x))dx = \int_a^b f(x)dx + \int_a^b g(x)dx$
- *Linear Functions*
  $\int_a^b (k_1 f(x) + k_2 g(x))dx = k_1 \int_a^b f(x)dx + k_2 \int_a^b g(x)dx$

Properties of Definite Integrals II

- *Intermediate Values*

  for $a \leq b \leq c$:
  $\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx$

- *Limit Reversibility*

  $\int_a^b f(x)dx = -\int_b^a f(x)dx$

Ex: $\int_0^1 x^2 dx = \int_0^{0.5} x^2 dx + \int_{0.5}^1 x^2 dx = \left(\frac{1}{3}x^3\right)_0^{0.5} + \left(\frac{1}{3}x^3\right)_{0.5}^1 =$
$\frac{1}{3}(0.125 - 0) + \frac{1}{3}(1 - 0.125) = \frac{1}{3}$

Indefinite Integrals

Indefinite Integrals

Definite integrals have limits for the integration (what we showed with a and b in the definition.) Indefinite integrals do not. Then, the antiderivative solution will have a constant, $+c$ added to them to account for this.

Definition 4 (Indefinite Integral).

The *indefinite integral* of $f(x)$ is given by:

$$\int f(x)dx := F(x) + c$$

Ex: $\int (1-3x)^{\frac{1}{2}} dx = -\frac{2}{9}(1-3x)^{\frac{3}{2}} + c$

Indefinite Integrals — Properties

- *Linearity (as for definite integrals):*
$$\int \Big[a\, f(x) + b\, g(x)\Big]\, dx \;=\; a \int f(x)\, dx \;+\; b \int g(x)\, dx$$
- *Antiderivative family:* If $F'(x) = f(x)$, then $\int f(x)\, dx = F(x) + C$; any two antiderivatives differ by a constant.
- *Constant function:* $\int c\, dx = cx + C$
- *Verification:* Differentiate your result; for definite integrals constants cancel:
$$\Big[F(x) + C\Big]_a^b = F(b) - F(a)$$

Special Cases

Special Integrals

- $\int x^a \, dx = \frac{1}{a+1} x^{a+1} + C$, where $a \neq -1$
- $\int \frac{1}{x-a} \, dx = \ln |x - a| + C$
- $\int e^{ax} \, dx = \frac{1}{a} e^{ax} + C$, where $a \neq 0$
- $\int a^x \, dx = \frac{1}{\ln a} a^x + C$, where $a > 0$ and $a \neq 1$

More Integration Techniques

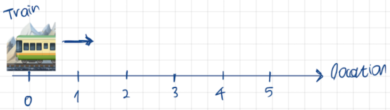
- For *Integration by Substitution*, see pages 218-219;
- For *Integration by Parts*, see pages 219-220-221;
- For *Multidimensional Integrals*, see Ch6.5 (pg 250) of J. Gill, 2006.

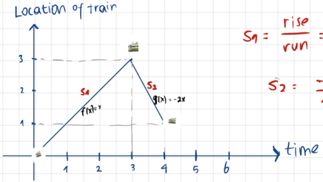
An Application: Inequality and the Gini Coefficient

- To be discussed in class.

Appendix

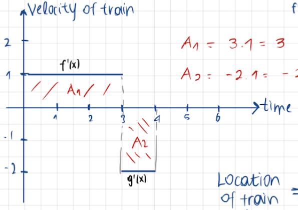
Areas, Integrals : Train Example





$S_1 = \frac{\text{rise}}{\text{run}} = \frac{3}{3} = 1 = f'(x)$

$S_2 = \frac{-2}{1} = -2 = g'(x)$



$A_1 = 3 \cdot 1 = 3 = \underbrace{\int_0^3 f'(x)\,dx}_{\text{the (Riemann) integral of } f'(x) \text{ between 0 and 3}} =$ Change in Location of train from $t=0$ to $t=3$

$A_2 = -2 \cdot 1 = -2 = \underbrace{\int_3^4 g'(x)\,dx}_{\text{the (Riemann) integral of } g'(x) \text{ between 3 and 4}} =$ Change in Location of train from $t=3$ to $t=4$

Location of train at $t=4$ $= A_1 + A_2$

$$= \int_0^3 f'(x)\,dx + \int_3^4 g'(x)\,dx$$